KAIZEN MATHS

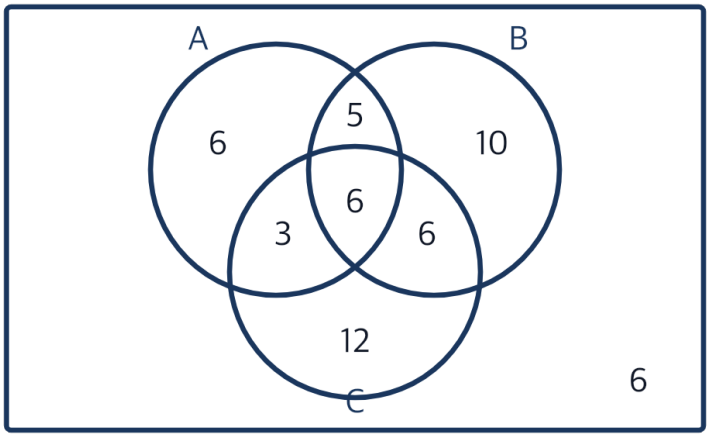
Lesson starter

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

SA1



Use the Venn diagram to find **n(A ∩ B)**.

STA-B8CE-L3 · SA

Show your working

KAIZEN MATHS

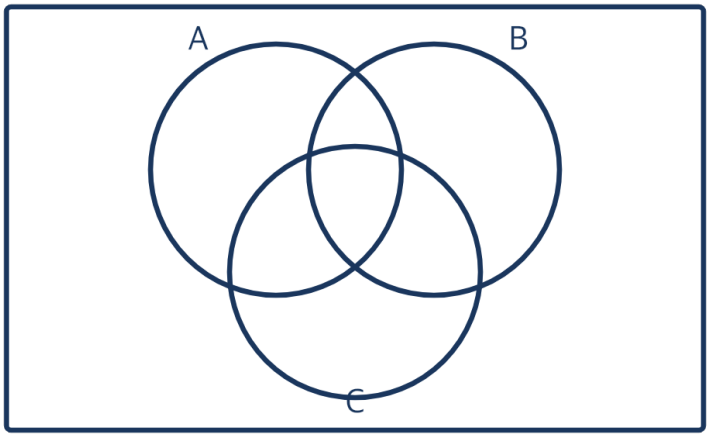
Lesson starter

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

SB1



In a survey of 39: n(A)=17, n(B)=23, n(C)=17, n(A ∩ B)=10, n(A ∩ C)=8, n(B ∩ C)=9, and n(A ∩ B ∩ C)=5. Find the number in neither set.

STA-B8CE-L3 · SB

Show your working

KAIZEN MATHS

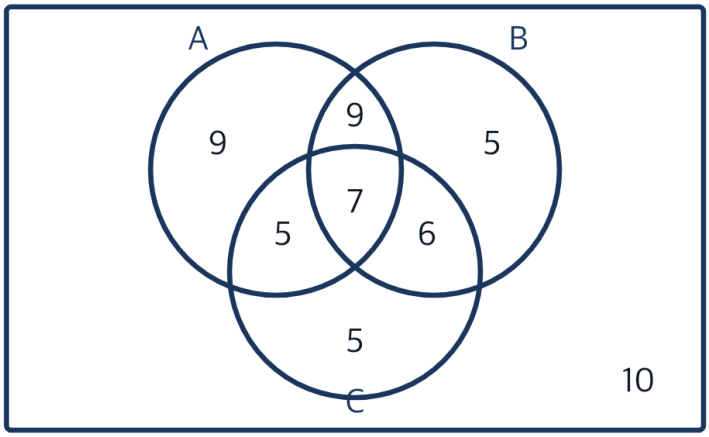
Lesson starter

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

SC1



A survey of 56 households used three sets: A=Laptop, B=Tablet, C=Phone. The completed diagram is shown. Find how many are in **exactly two** of the sets.

STA-B8CE-L3 · SC

Show your working

KAIZEN MATHS

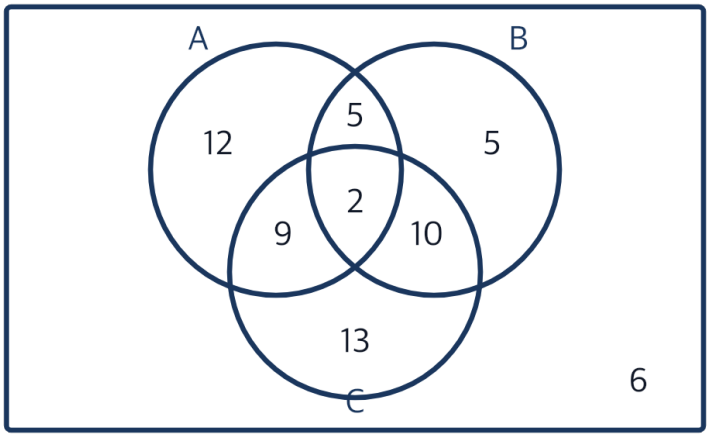
Lesson starter

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

SD1



Use the Venn diagram to find **at least two sets**.

STA-B8CE-L3 · SD

Show your working

KAIZEN MATHS

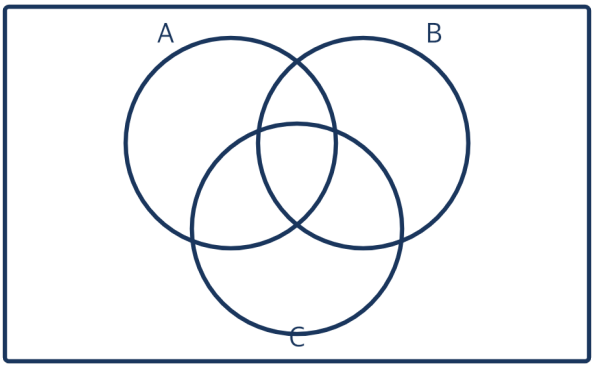
Exit ticket

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

EA1



In a survey of 60: n(A)=29, n(B)=25, n(C)=31, n(A ∩ B)=10, n(A ∩ C)=9, n(B ∩ C)=13, and n(A ∩ B ∩ C)=3. Find the number in neither set.

Show your working

KAIZEN MATHS

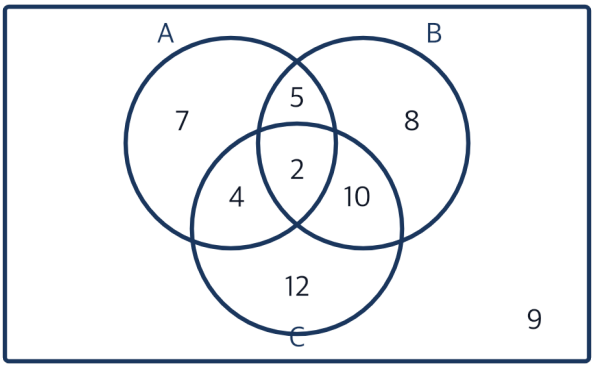
Exit ticket

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

EB1



A survey of 57 students used three sets: A=Football, B=Cricket, C=Netball. The completed diagram is shown. Find how many are in **exactly two** of the sets.

Show your working

KAIZEN MATHS

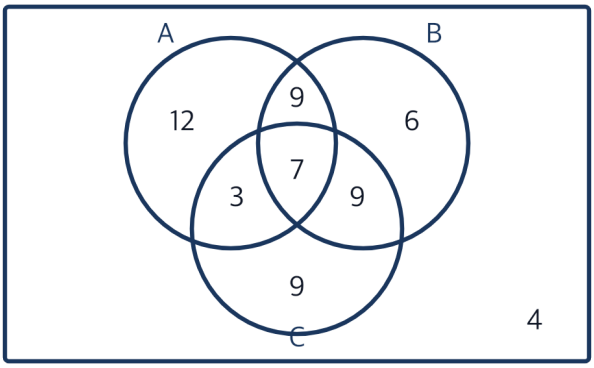
Exit ticket

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

EC1



Use the Venn diagram to find **n(A ∩ B)**.

Show your working

KAIZEN MATHS

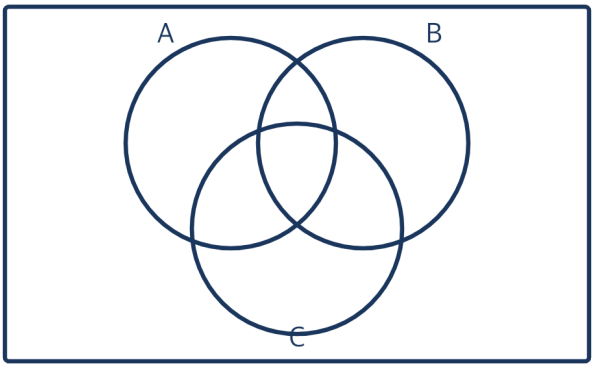
Exit ticket

Venn Diagrams and Set Probability

Three-Set Problems

Name _____

EA1



In a survey of 60: n(A)=29, n(B)=25, n(C)=31, n(A ∩ B)=10, n(A ∩ C)=9, n(B ∩ C)=13, and n(A ∩ B ∩ C)=3. Find the number in neither set.

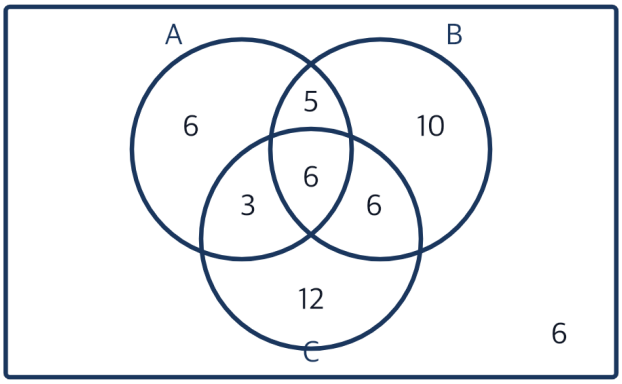
Show your working

Teacher answers & working

Venn Diagrams and Set Probability · Three-Set Problems

STA-B8CE-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Use the Venn diagram to find **n(A ∩ B)**.

Answer:

11

Working:

A ∩ B includes A ∩ B ∩ C′ and A ∩ B ∩ C.

This means the A and B only region, plus the centre.

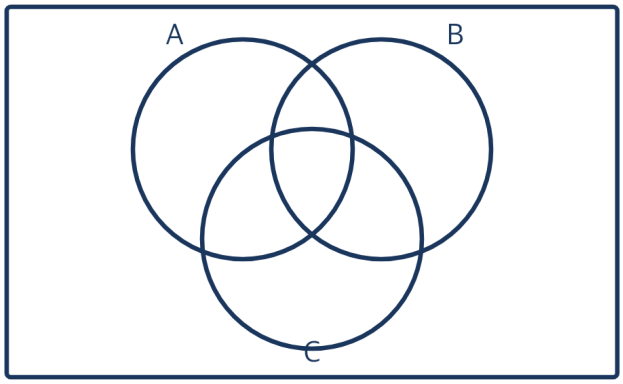
$n(A \cap B) = 5 + 6 = 11.$

Teacher answers & working

Venn Diagrams and Set Probability · Three-Set Problems

STA-B8CE-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



In a survey of 39: n(A)=17, n(B)=23, n(C)=17, n(A ∩ B)=10, n(A ∩ C)=8, n(B ∩ C)=9, and n(A ∩ B ∩ C)=5. Find the number in neither set.

Answer:

Neither = 4

Working:

Start in the centre: n(A ∩ B ∩ C) = 5.

$n(A \cap B \cap C') = n(A \cap B) - n(A \cap B \cap C) = 10 - 5 = 5.$

This is A and B only: in A and B, but not in C.

$n(A \cap C \cap B') = n(A \cap C) - n(A \cap B \cap C) = 8 - 5 = 3.$

This is A and C only: in A and C, but not in B.

$n(B \cap C \cap A') = n(B \cap C) - n(A \cap B \cap C) = 9 - 5 = 4.$

This is B and C only: in B and C, but not in A.

$n(A \cap B' \cap C') = n(A) - (5 + 3 + 5) = 17 - (5 + 3 + 5) = 4.$

This is A only: in A but not in B or C.

$n(A' \cap B \cap C') = n(B) - (5 + 4 + 5) = 23 - (5 + 4 + 5) = 9.$

This is B only: in B but not in A or C.

$n(A' \cap B' \cap C) = n(C) - (3 + 4 + 5) = 17 - (3 + 4 + 5) = 5.$

This is C only: in C but not in A or B.

$n((A \cup B \cup C)') = 39 - 35 = 4.$

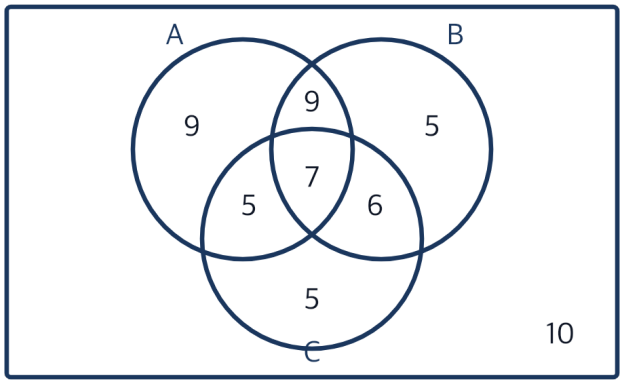
This means outside all three sets: neither A, B, nor C.

Teacher answers & working

Venn Diagrams and Set Probability · Three-Set Problems

STA-B8CE-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



A survey of 56 households used three sets: A=Laptop, B=Tablet, C=Phone. The completed diagram is shown. Find how many are in **exactly two** of the sets.

Answer:

20

Working:

Exactly two sets means the pairwise-only regions, not the centre A ∩ B ∩ C.

$n(A \cap B \cap C') + n(A \cap C \cap B') + n(B \cap C \cap A') = 9 + 5 + 6.$

This adds A and B only, A and C only, and B and C only.

Exactly two = 20.

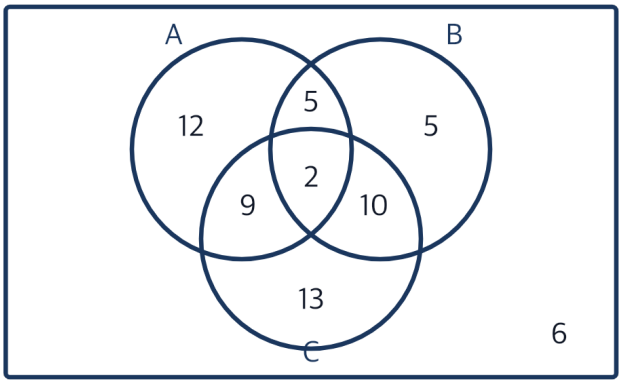
Check: n(A ∪ B ∪ C) = 46.

Teacher answers & working

Venn Diagrams and Set Probability · Three-Set Problems

STA-B8CE-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Use the Venn diagram to find **at least two sets**.

Answer:

26

Working:

$A \cap B \cap C'$, $A \cap C \cap B'$, $B \cap C \cap A'$, $A \cap B \cap C$

These are the three pair-only regions plus the centre.

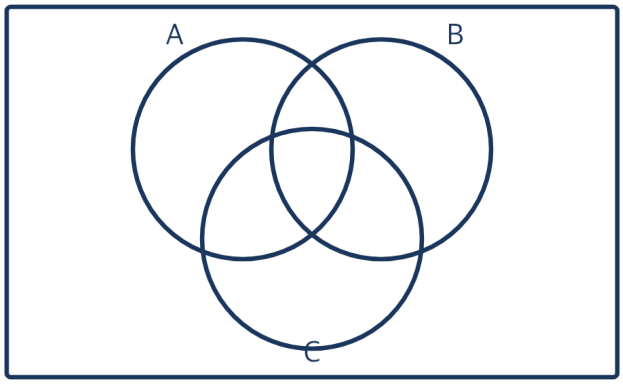
At least two sets = 5 + 9 + 10 + 2 = 26.

Teacher answers & working

Venn Diagrams and Set Probability · Three-Set Problems

STA-B8CE-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



In a survey of 60: n(A)=29, n(B)=25, n(C)=31, n(A ∩ B)=10, n(A ∩ C)=9, n(B ∩ C)=13, and n(A ∩ B ∩ C)=3. Find the number in neither set.

Answer:

Neither = 4

Working:

Start in the centre: n(A ∩ B ∩ C) = 3.

$n(A \cap B \cap C') = n(A \cap B) - n(A \cap B \cap C) = 10 - 3 = 7.$

This is A and B only: in A and B, but not in C.

$n(A \cap C \cap B') = n(A \cap C) - n(A \cap B \cap C) = 9 - 3 = 6.$

This is A and C only: in A and C, but not in B.

$n(B \cap C \cap A') = n(B \cap C) - n(A \cap B \cap C) = 13 - 3 = 10.$

This is B and C only: in B and C, but not in A.

$n(A \cap B' \cap C') = n(A) - (7 + 6 + 3) = 29 - (7 + 6 + 3) = 13.$

This is A only: in A but not in B or C.

$n(A' \cap B \cap C') = n(B) - (7 + 10 + 3) = 25 - (7 + 10 + 3) = 5.$

This is B only: in B but not in A or C.

$n(A' \cap B' \cap C) = n(C) - (6 + 10 + 3) = 31 - (6 + 10 + 3) = 12.$

This is C only: in C but not in A or B.

$n((A \cup B \cup C)') = 60 - 56 = 4.$

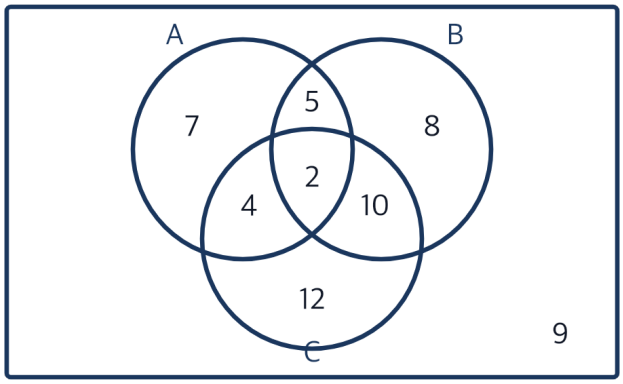
This means outside all three sets: neither A, B, nor C.

Teacher answers & working

Venn Diagrams and Set Probability · Three-Set Problems

STA-B8CE-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



A survey of 57 students used three sets: A=Football, B=Cricket, C=Netball. The completed diagram is shown. Find how many are in **exactly two** of the sets.

Answer:

19

Working:

Exactly two sets means the pairwise-only regions, not the centre A ∩ B ∩ C.

$n(A \cap B \cap C') + n(A \cap C \cap B') + n(B \cap C \cap A') = 5 + 4 + 10.$

This adds A and B only, A and C only, and B and C only.

Exactly two = 19.

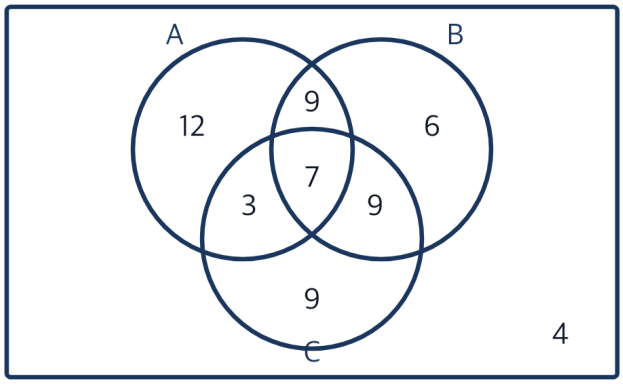
Check: n(A ∪ B ∪ C) = 48.

Teacher answers & working

Venn Diagrams and Set Probability · Three-Set Problems

STA-B8CE-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Use the Venn diagram to find **n(A ∩ B)**.

Answer:

16

Working:

A ∩ B includes A ∩ B ∩ C′ and A ∩ B ∩ C.

This means the A and B only region, plus the centre.

$n(A \cap B) = 9 + 7 = 16$.